Step 1 (v2): Decision Audit and Freeze-Ready Parameters

Particle Bed Reactor Stability Simulator

SwampHacks XII Project

Everything that can be decided before writing code

How to read this. Each of the ten decisions from Step 1 was checked against two tests: **(1) Is it the best available choice?** and **(2) Could it lead to hardcoding**, meaning a result decided by our choices rather than by the physics? Decisions marked ✓ passed unchanged. Decisions marked ✓ mod passed after a modification, which is explained. Section 4 is the final parameter table; Section 6 lists what can only be settled once code exists.

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1 Results at a Glance

#	Decision	Status	What changed
D1	Inlet 200 K; 8.0 MPa at outlet	✓ mod	Added a sensitivity case with 8.0 MPa at the inlet, because the source does not say where it applies.
D2	Mass flow by energy balance	✓ mod	Became a *rule* computed by the property module, not a typed number; throttling keeps outlet temperature fixed.
D3	Property sources	✓ mod	Transport above 1000 K now uses specific, verified NASA coefficients; smooth blending at 1000 K plus a continuity test; dissociation and para-hydrogen quantified.
D4	Correlations	✓ mod	Validity ranges checked: pressure drop valid, heat transfer partly outside range; added a sensitivity correlation and a rule to flag out-of-range points.
D5	Fixed mass flow, fixed outlet pressure	✓	Confirmed; throttling rule added under D2.
D6	Channel friction	✓	Petukhov correlation chosen; Reynolds numbers confirmed in range.
D7	Fuel heat capacity	✓ mod	Earlier claim corrected: only the *static* stability boundary is independent of heat capacity. Sourced proxy value added.
D8	Hot-frit-dominated design	✓	Confirmed as a rule computed in code.
D9	Study ranges	✓ mod	Power range extended to 5% so it brackets the NASA threshold; optimizer penalty weight replaced by a Pareto front; "weak frit" threshold replaced by a continuous sweep.
D10	Validation criteria	✓ mod	Rewritten per target, using the paper's own inputs; added a pre-registration rule.

Three decisions could have led to hardcoding and were fixed: a hand-picked optimizer penalty weight (D9), a hand-picked "weak frit" cutoff (D9), and a mass flow typed in rather than derived (D2). Two decisions could have introduced *artificial physics*: a kink in properties at 1000 K (D3) and an overstated claim about heat capacity (D7). Both were fixed.

2 Decision Audit

2.1 D1: Inlet temperature and operating pressure ✓ mod

Decision: inlet 200 K; 8.0 MPa at the hot channel exit.

Best choice? No source states these for the fuel element. Both come from the other reactor in the same paper [T19], which is the closest consistent source. Ji et al. 2020 [J20] treats inlet temperature as a parameter it varies, so 200 K is a design choice, not a fixed fact.

Hardcoding risk: low, since the values are fixed before any run. But the paper calls 8.0 MPa the operating pressure "for the whole reactor" without saying where it applies. With a pressure drop of about 1.2 MPa, that ambiguity moves gas density by roughly 15%.

Modification: keep 8.0 MPa at the outlet as primary. Run every validation case a second time with 8.0 MPa at the inlet and report both. The spread becomes an honest uncertainty band instead of a hidden choice.

2.2 D2: Mass flow from energy balance ✓ mod

Decision: keep geometry and 30 MW/L; set mass flow so the outlet reaches 3000 K.

Best choice? Yes. Power density is the variable in which the NASA threshold [K93] and the throttle range [SNTP] are expressed, and 3000 K is the SNTP design target [SNTP].

Hardcoding risk: if the mass flow were typed in as 0.447 kg/s from our hand calculation, it would not match the code's own property model exactly, and there would be a temptation to adjust it later.

Modification: mass flow is a **rule**, computed by the property module:

$$\dot{m} = \frac{P_{\text{element}}}{h(T_{\text{target}}, p_{\text{out}}) - h(T_{\text{in}}, p_{\text{out}})}, \qquad T_{\text{target}} = 3000 \text{ K}$$

Throttling rule: at reduced power, $\dot{m}$ scales with power so the outlet stays at 3000 K. This matches the NASA scenario of operating at design exit temperature but reduced power [K93]. The published 0.396 kg/s is kept as a sensitivity case and is the input for validation runs.

2.3 D3: Hydrogen properties ✓ mod

Decision: CoolProp below 1000 K; NIST-JANAF above; NASA transport coefficients above 1000 K.

Best choice? Yes, with refinements:

- **CoolProp below 1000 K** is the right choice there because density effects matter at 200 K and about 9 MPa, which is exactly where the cold frit, the stabilizing element, operates.
- **NASA transport coefficients** [SV95] were retrieved and checked. For H_2, viscosity and conductivity coefficients are given for 200–1000 K and 1000–5000 K. Our evaluation shows the two intervals meet continuously at 1000 K (viscosity 2.056×10^{-5} Pa·s on both sides). These are dilute-gas values, appropriate at high temperature.

- **NIST-JANAF** gives ideal-gas c_p and enthalpy to 6000 K [JANAF].

Artificial-physics risk: switching sources at 1000 K can create a small jump or kink in properties. A stability analysis takes derivatives, so a kink could create a false instability.

Modification:

1. Blend CoolProp and high-temperature values smoothly over 950–1050 K.
2. Interpolate JANAF tables with a smooth monotone method (PCHIP), not straight lines, so c_p has no steps.
3. Align enthalpy reference values at 1000 K (a constant offset, not a fit).
4. **Verification test:** property values and slopes must be continuous across 900–1100 K.
5. Above 1000 K, density from the ideal-gas law. **Test:** compare with CoolProp at 1000 K and both pressures; report the difference.

Two effects quantified and accepted as limitations:

- **Dissociation.** From JANAF data for atomic hydrogen at 8 MPa, the hydrogen-atom mole fraction is about 0.9% at 2800 K, 1.8% at 3000 K, 3.1% at 3200 K, and 6.4% at 3500 K. At 3000 K, dissociation absorbs roughly 4% of the energy. Tier 1 neglects it; it would lower outlet temperature by roughly 100 K, which explains part (not all) of the energy discrepancy found in Step 1. **Tier 3:** add equilibrium dissociation using the JANAF tables.
- **Para-hydrogen.** Real propellant comes from liquid hydrogen, which is para-hydrogen; one early PBR transient analysis treats the coolant as para-hydrogen [TR90]. Tier 1 uses normal hydrogen (consistent with JANAF); run CoolProp's para-hydrogen below 1000 K as a sensitivity case.

2.4 D4: Correlations ✓ mod

Decision: KTA 3102.3 bed pressure drop, Ergun frits, KTA 3102.2 heat transfer (all as in [T19]).

Validity ranges checked:

- **KTA pressure drop:** valid for $0.36 < \varepsilon < 0.42$ and $10 < \mathrm{Re}_m < 10^5$, with no wall-channeling effects [RG22]. Our bed porosity is 0.38 ✓.
- **KTA heat transfer:** valid for $100 < \mathrm{Re} < 10^5$, $0.36 < \varepsilon < 0.42$, bed diameter at least 20 particle diameters, and bed height above 4 particle diameters [HT].

Our operating Reynolds numbers (computed with NASA viscosities, mass flow from D2):

Power	Re, cold face	Re_m, cold face	Re, hot face	Re_m, hot face
100%	268	433	50	81
20%	54	87	10	16
5%	13	22	2.5	4.1

Findings:

- KTA pressure drop is **in range** at 100% and 20% power, and **out of range** at the hot face below roughly 15% power.
- KTA heat transfer is **out of range** on the hot side even at full power ($\mathrm{Re} < 100$), and the bed is only about 12 particles thick (below 20).
- The bed is thin, so wall effects are a concern. Measurements found the KTA pressure drop matched well for bed-to-particle ratios down to about 6 in cylindrical beds [HK12], which supports its use here.

Why heat transfer being out of range is acceptable: in steady state, each slice's gas outlet temperature is fixed by its energy balance (power divided by flow times heat capacity), independent of the heat transfer coefficient. The heat transfer coefficient mainly sets how much hotter the fuel is than the gas. So it affects peak fuel temperature, not the flow split that drives instability.

Modifications:

1. Keep KTA for comparability with [T19].
2. Add Wakao–Kaguei ($\mathrm{Nu} = 2 + 1.1\,\mathrm{Pr}^{1/3}\mathrm{Re}^{0.6}$) as the heat transfer sensitivity case and Ergun as the pressure-drop sensitivity case.
3. **Rule:** the code flags every result computed outside a correlation's validity range, and reports Ergun results alongside for those points (the Ergun viscous term remains well founded at low Reynolds number).

2.5 D5: Boundary condition ✓

Decision: fixed total mass flow, fixed outlet pressure.

Best choice? Yes. It matches the paper's simulation setup (mass flow specified at the inlet) [T19], and the instability of interest, flow redistributing among slices inside one element, still appears under fixed total flow. With the throttling rule from D2, it also reproduces the NASA operating scenario. **Hardcoding risk:** none.

2.6 D6: Hot channel and plenum friction ✓

Decision: Petukhov smooth-tube correlation, $f = (0.790 \ln \mathrm{Re} - 1.64)^{-2}$, valid for about $3000 \leq \mathrm{Re} \leq 5 \times 10^6$ [INC].

Check: hot channel Reynolds number about 2.9×10^5 at full power and 5.9×10^4 at 20%; inlet plenum about 6×10^5. All in range ✓.

Momentum with mass transfer: the hot channel gains mass along its length and the plenum loses it. Use the standard manifold momentum balance in which gas entering the channel carries no axial momentum, as in the manifold theory cited by [T19] [W11]. State that wall suction and injection effects on friction are neglected.

2.7 D7: Fuel heat capacity ✓ mod

Earlier claim: heat capacity cannot change whether a design is stable.

Correction: that is only partly true. In the model, fuel temperatures are the time-varying quantities, so the linearized system is $\dot{\mathbf{x}} = C^{-1}G\,\mathbf{x}$, with C a diagonal matrix of positive heat capacities.

- **Static instability** (a real eigenvalue crossing zero, the runaway type) occurs where $\det G = 0$. Because $\det(C^{-1}G) = \det(C^{-1})\det(G)$, this boundary is **independent of heat capacity** ✓.
- **Oscillatory instability** (complex eigenvalues crossing zero) **can** depend on how heat capacity differs between slices, for example because c_p rises with temperature.

Value: the particle has a UC_2 kernel, pyrocarbon layers, and a ZrC or NbC coating [SNTP]. Use ZrC as a proxy, from NIST-JANAF [JZrC]: c_p = 53.388, 57.392, and 60.906 J/(mol·K) at 1000, 2000, and 3000 K, which is about 517, 556, and 590 J/(kg·K). Take density from a ZrC property review [KA13].

Modification: report static and oscillatory boundaries separately; run heat capacity at ±50% and with constant versus temperature-dependent c_p; state that any oscillatory result depends on this proxy.

2.8 D8: Hot-frit-dominated design ✓

Decision: construct it from the paper's cold-frit-dominated element by swapping which frit carries the larger pressure-drop share at nominal conditions.

Best choice? Yes; the paper does not publish these values [T19], and a rule tied to the paper's own design is the most defensible construction.

Hardcoding risk: avoided because the frit values are *computed in code by the rule* before any stability run. If our construction turns out stable, that is reported as a finding, not fixed.

Precise rule: "frit share" = pressure drop across that frit divided by total element pressure drop, at nominal steady state. Adjust hot frit pore size until its share equals the cold frit's original share; adjust cold frit pore size until its share equals the hot frit's original share. Porosities unchanged.

2.9 D9: Study ranges ✓ mod

Three problems found:

1. **The power range missed the NASA threshold.** At 30 MW/L nominal, 20% power is 6 MW/L, above the roughly 4 MW/L threshold [K93]. **Fix:** sweep 5%–100% (1.5–30 MW/L). Label 20%–100% as the SNTP operating range [SNTP], and flag points where correlations are out of range (D4).
2. **A hand-picked penalty weight would decide the optimizer's answer.** Choosing how much to penalize pressure drop effectively chooses the winning design. **Fix:** report a Pareto front (the set of best trade-offs) between peak fuel temperature and total pressure drop, with a constraint that the design stays stable across the 20%–100% throttle range. That stability

constraint is also what distinguishes this optimizer from the earlier genetic-algorithm study [J19b].

3. **A hand-picked "weak frit" cutoff** (under 5% of pressure drop) would decide where the line falls. **Fix:** sweep frit share continuously from near zero to the paper's design value and report the full curve.

Unchanged: 8 axial frit zones; multipliers 0.25–4 on a log scale; fuel limit 2800 K by default with 3100 K and 3500 K options [SNTP]; flow patterns Z and U.

2.10 D10: Validation criteria

✓ mod

Modification: criteria are written per target, use the paper's inputs exactly, and are committed to Git before any validation run (Section 5).

3 Anti-Hardcoding Protocol

These rules apply from the moment `params.py` is written.

1. **Freeze before validation.** `params.py` and the validation criteria are committed to Git before any validation run. The commit hash is recorded in every result file.
2. **No tuning to targets.** No parameter is changed to improve agreement with a validation target.
3. **Literature inputs for validation.** Validation runs use the source paper's inputs exactly, even where Step 1 found them inconsistent.
4. **Rules, not numbers, for derived values.** Mass flow (D2), the hot-frit-dominated design (D8), and grid size are computed by stated rules in code.
5. **Versioned changes.** If a genuine error is found after validation, fix it as a new model version, rerun everything, and report both versions.
6. **Report all outcomes,** including failed validation criteria.
7. **Synthetic data stays separate** and is never reported as a result.
8. **The app always runs the solver live** on the judge's inputs.

4 Freeze-Ready Parameter Table

Status key: **S** = from a source; **A** = assumption (labeled); **R** = computed in code by a stated rule; **D** = design choice for the study.

4.1 Nominal fuel element

Parameter	Value or rule	Status	Source / note
Bed length	0.6392 m	S	[T19] Table 5
Hot channel radius	0.02129 m	S	[T19]
Hot frit thickness	0.003 m	S	[T19]
Bed thickness	0.00615 m	S	[T19]
Cold frit thickness	0.002 m	S	[T19]
Inlet plenum thickness	0.003 m	S	[T19]
Particle diameter	0.0005 m	S	[T19], [SNTP]
Bed porosity	0.38	S	[T19]
Cold frit pore size / porosity	2 μm / 0.4	S	[T19] (cold-frit-dominated)
Hot frit pore size / porosity	0.1 mm / 0.5	S	[T19]
Power density (bed volume)	30 MW/L	S	[T19]; gives 20.28 MW
Axial power shape	Uniform	S	[T19]; non-uniform is future work
Inlet temperature	200 K	A	From [T19]'s other case (D1)
Pressure reference	8.0 MPa at outlet; inlet-referenced sensitivity	A	D1
Target outlet temperature	3000 K	S	[SNTP], [T19]
Mass flow	Energy-balance rule (D2)	R	$\approx$0.447 kg/s by hand estimate
Mass flow, sensitivity	0.396 kg/s	S	[T19]
Throttling	$\dot{m} \propto$ power	D	Keeps 3000 K outlet (D2)
Hot-frit-dominated frits	Swap-shares rule (D8)	R	

4.2 Validation case (low-power-density reactor)

All values exactly as in [T19]: core radius 507 mm; length 1200 mm (also 1000 and 800 mm); hot channel radius 100 mm; hot frit 3 mm; bed 281 mm; cold frit 2 mm; plenum 10 mm; particles 6 mm; frit pore size 4 mm; porosity 0.38 (bed), 0.1 (cold frit), 0.2 (hot frit); 1000 MW uniform; inlet 200 K; 21, 17.5, and 14 kg/s for the three heights; 8.0 MPa. Frit-dominance variants: cold frit pore 10 μm with porosity 0.2; hot frit pore 400 μm.

4.3 Hydrogen properties

Item	Choice
Species	Normal hydrogen; para-hydrogen sensitivity below 1000 K
200–1000 K	CoolProp real-gas properties [CP]
Above 1000 K, thermodynamic	NIST-JANAF H_2 table with PCHIP interpolation [JANAF]
Above 1000 K, density	Ideal gas
Above 1000 K, transport	NASA coefficients below [SV95]
Junction	Smooth blend over 950–1050 K; enthalpy offset aligned at 1000 K
Dissociation	Neglected in Tier 1; Tier 3 equilibrium from JANAF

NASA H_2 transport coefficients [SV95]. Form: $\ln\eta$, $\ln\lambda = A\ln T + B/T + C/T^2 + D$, with η in micropoise ($1\ \mu\text{P} = 10^{-7}$ Pa·s) and λ in μW/(cm·K) ($1 = 10^{-4}$ W/(m·K)).

Property	Range (K)	A	B	C	D
Viscosity	200–1000	0.74553182	43.555109	−3257.9340	0.13556243
Viscosity	1000–5000	0.96730605	679.31897	−210251.79	−1.8251697
Conductivity	200–1000	1.0240124	297.09752	−31396.363	1.0560824
Conductivity	1000–5000	1.0611992	258.85783	6316.3191	0.79973205

Spot values from these coefficients: viscosity 6.82, 8.97, 20.56, 33.51, and 45.60 $\times 10^{-6}$ Pa·s at 200, 300, 1000, 2000, and 3000 K. **Before use, copy the coefficients again directly from the PDF** to rule out transcription errors.

4.4 Correlations

Use	Primary	Sensitivity / notes
Bed pressure drop	KTA 3102.3 [T19]	Ergun; flag $\mathrm{Re}_m < 10$
Frit pressure drop	Ergun with pore size and frit porosity [T19]	
Fuel-to-gas heat transfer	KTA 3102.2 [T19]	Wakao–Kaguei; flag $\mathrm{Re} < 100$
Interfacial area	$6(1-\varepsilon)/d_p$ [T19]	
Channel and plenum friction	Petukhov [INC]	Range 3000–5×10^6
Channel momentum	Manifold balance, injected gas has no axial momentum [W11]	

4.5 Fuel, limits, and study settings

Item	Value	Source / note
Fuel heat capacity	ZrC proxy: 517–590 J/(kg·K), 1000–3000 K	[JZrC]; ±50% sensitivity
Fuel density	ZrC, from review	[KA13]
Fuel temperature limit	2800 K default; 3100 and 3500 K options	[SNTP]
Power sweep	5%–100% (1.5–30 MW/L)	D9; 20%–100% labeled operational
Frit share sweep	Continuous, near zero to design value	D9
Flow patterns	Z (baseline), U	[T19]
Optimizer zones / bounds	8 zones; multipliers 0.25–4 (log)	D9
Optimizer objectives	Pareto: peak fuel temperature vs. pressure drop; stable over 20%–100%	D9

5 Validation Criteria (Pre-Registered)

All runs use the paper's stated inputs; the inlet-referenced pressure variant is run alongside.

ID	Target	Pass criterion
V1	Validation case, Z: Δp = 872.4 kPa [T19]	Within ±25%
V2	Z vs. U: T_{max} falls (7121.3 to 4833.0 K), Δp nearly unchanged [T19]	U gives lower peak temperature than Z; Δp changes by less than 2%
V3	Core heights 80/100/120 cm [T19]	Δp within ±25% at each; nonuniformity ratio β decreases as height decreases
V4	Validation case frit variants [T19]	Both variants lower peak temperature vs. baseline
V5	Fuel element, cold-frit-dominated, 0.396 kg/s: Δp = 1189.2 kPa [T19]	Within ±25%; steady solution is stable
V6	Fuel element, hot-frit-dominated (D8 rule) [T19]	Model predicts instability or no steady solution below 8000 K
V7	No cold frit, design exit temperature [K93]	Static instability appears below a power density within a factor of 3 of 4 MW/L
V8	Strong cold frit [K93]	The V7 instability disappears
V9	Ledinegg check	Eigenvalue onset and pressure-drop-slope onset agree within one sweep step

Expected disagreements to report, not fix: absolute temperatures (energy discrepancy, Step 1); 1D model vs. 2D CFD; parallel-passage limitation at low Reynolds number [K93].

6 What Can Only Be Settled With Code

These are not open decisions; each has a fixed rule. They need code to evaluate.

Item	Rule
Exact nominal mass flow	D2 formula using the property module
Hot-frit-dominated frit values	D8 swap-shares rule
Number of slices and radial steps	Start at 20 slices and 20 radial steps; double until peak fuel temperature and total Δp change by less than 1%
Property continuity at 1000 K	Values and slopes continuous over 900–1100 K
Ideal-gas error at 1000 K	Compare with CoolProp density; report
Correlation range flags	Computed per run (D4)

7 Remaining Non-Code Items

- ☐ Get Ji et al. 2019 [J19b] and 2020 [J20] and Ludewig et al. 1996 through UF Libraries; record any inlet conditions or frit profiles they give. If they state the fuel element's inlet conditions, replace the D1 assumption *before* freezing.
- ☐ Get the ZrC density from [KA13].
- ☐ Download the NIST-JANAF H_2 and H tables as data files.
- ☐ Copy the NASA coefficients directly from [SV95] and double-check them.
- ☐ Confirm the Wakao–Kaguei validity range in its source.
- ☐ Add Bussard and DeLauer [BD58] to the sources after confirming the edition Kerrebrock cites.
- ☐ Ask a UF Nuclear Engineering faculty member to review D1, D2, D8, and the criteria in Section 5.

Ready to code. Once the items above are done, every input, rule, criterion, and protocol is fixed. The next step is writing `params.py` and `properties.py` directly from this document, then the verification tests, then committing all of it before the first validation run.

Sources

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